



PAPER ID-311822

Subject Code: BMBAI102

Roll No:

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

MBA-INT
(SEM I) THEORY EXAMINATION 2024-25
BUSINESS MATHEMATICS

TIME: 3 HRS

M.MARKS: 70

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt all questions in brief.

2 x 07 = 14

Q no.	Question	CO	Level
a.	Explain the Law of Indices.	1	K2
b.	Define Ven Diagram.	2	K1
c.	Explain set operation.	2	K2
d.	Name the types of matrices.	3	K1
e.	Given the matrices $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ $B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$ Find i) $A+B$ ii) $A-B$	3	K1
f.	List the types of interest rates.	4	K1
g.	Explain the uses of probability in various fields.	5	K2

SECTION B

2. Attempt any three of the following:

7 x 3 = 21

a.	Find the sum of the first 10 terms of an arithmetic progression where the first term is 3 and the common difference is 5.	1	K1
b.	Evaluate the effectiveness of using Set Theory in business decision-making processes.	2	K5
c.	Solve the system of linear equations using the Matrix Inverse Method: $2x+y=5$ $x-y=2$	3	K6
d.	Distinguish between an ordinary annuity and an annuity due, and explain their significance.	4	K4
e.	Evaluate the effectiveness of using Bayes' theorem in medical diagnosis compared to traditional probability methods.	5	K5

SECTION C

3. Attempt any one part of the following:

07 x 1 = 07

a.	Examine a linear function and a quadratic function, and provide examples of each.	1	K4
b.	A farmer wants to build a rectangular pen against a barn using 80 meters of fencing. Determine what dimensions will maximize the area of the pen?	1	K5

4. Attempt any one part of the following:

07 x 1 = 07

a.	Define the following terms related to Set Theory: • Union • Intersection • Complement • Power set	2	K1
b.	Evaluate the truth of De Morgan's Laws using the sets: $X = \{2,4,6,8\}$ and $Y = \{1,3,5,7\}$ in the universal set	2	K5

Roll No:

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

**MBA-INT
(SEM I) THEORY EXAMINATION 2024-25
BUSINESS MATHEMATICS**

TIME: 3 HRS

M.MARKS: 70

	$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. Solve $(A \cup B)^c = A^c \cap B^c$ $(A \cap B)^c = A^c \cup B^c$		
--	--	--	--

5. Attempt any one part of the following: 07 x 1 = 07

a.	Solve the system of linear equations using Cramer's Rule: $2x + 3y = 5$ $4x + y = 3$	3	K6
b.	Find the inverse of the matrix $A = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$.	3	K6

6. Attempt any one part of the following: 07 x 1 = 07

a.	Find the present value of an ordinary annuity with annual payments of ₹1000 for 5 years at an interest rate of 4%, compounded annually.	4	K1
b.	Discuss the valuation of a simple loan and a debenture, and explain the differences between them.	4	6

7. Attempt any one part of the following: 07 x 1 = 07

a.	Explain the addition theorem of probability and provide an example.	5	K5
b.	A company produces light bulbs, and the probability that a light bulb is defective is 0.02. If a customer buys 3 light bulbs, what is the probability that all 3 light bulbs are defective? https://www.aktuonline.com	5	K1